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1 Three proofs of the Kunen Inconsistency.

Some facts: In this section, let $j\colon V \to V$ be a non-trivial elementary embedding.

- j maps ordinals to ordinals, since being an ordinal is first-order definable.
- Define $\mathrm{crit}(j) = \kappa$ as the least ordinal moved by j. Why does such an ordinal exist? Assume no ordinal is moved by j, then the rank of a set is also fixed, by a rank induction we get $j \restriction V_\alpha = \mathrm{id}$ for all ordinal α, which leads to a contradiction that j is non-trivial. Now we have a class of ordinals that moved by j, namely $\{\alpha \in \mathrm{Ord}\colon j(\alpha) \neq \alpha\}$. But it is not necessarily a set, we take some α such that $j(\alpha) \neq \alpha$ and then consider the set $A := \{\beta < \alpha + 1 : j(\beta) \neq \beta\}$, let $\kappa := \min A$. (**Caution:** Here we implicitly used Separation with a class parameter j, which is not an instance of the Separation schema of ordinary ZFC. Thus, the argument requires some care regarding its formalization. However, we can work in NBG, which is equiconsistent with ZFC.[1]) But again, why must $j(\kappa) > \kappa$, rather than $j(\kappa) > \kappa$? Suppose $j(\kappa) < \kappa$ then $\forall \alpha < \kappa, j(\alpha) = \alpha$. Consider $j(j(\kappa))$, since j is elementary so $j(\kappa) < \kappa$ implies $j(j(\kappa)) < j(\kappa)$ but on the other hand $j(j(\kappa)) = j(\kappa)$ as κ is the least ordinal that moved by j. Contradiction!
- κ is a cardinal. Assume to the contract, let $\overline{\overline{\kappa}} = \lambda < \kappa$ and let $f\colon \lambda \to \kappa$ be a bijection. By elementarity, $j(f)$ is a bijection from $j(\lambda) = \lambda$ to $j(\kappa)$. But $j(\kappa) = j(f)''\lambda = f''\lambda = \kappa$, contradiction!
- **Reinhardt Cardinal:** a cardinal κ is called a Reinhardt cardinal if there is a non-trivial elementary embedding $j\colon V \to V$ and $\mathrm{crit}(j) = \kappa$.
- **Critical sequence:** let $\mathrm{crit}(j) = \kappa = \kappa_0$ and $\kappa_{\alpha+1} = j(\kappa_\alpha)$ for all $\alpha < \omega$, set $\lambda = \sup_{\alpha<\omega} \kappa_\alpha$.
  - λ is a fixed point of j : $j(\lambda) = j(\sup_{\alpha<\omega} \kappa_\alpha) = \sup_{\alpha<\omega} j(\kappa_\alpha) = \lambda$.
  - pointwise image of $\lambda := j''\lambda = \{j(\alpha) : \alpha < \lambda\}$, for $\beta < \lambda$, there exist some $n < \omega$ such that $\beta < \kappa_n$ hence $j(\beta) < \kappa_{n+1} < \lambda$, thus $j''\lambda \subseteq \lambda$.
  - $\kappa \notin j''\lambda : \forall \alpha < \kappa, j(\alpha) = \alpha < \kappa; \forall \alpha \geq \kappa, j(\alpha) > \kappa$. Together with the fact one line above we have $j''\lambda \subsetneq \lambda$.

1.1 Kunen's original proof

Definition 1.1. Let λ be an ordinal, an ω-Jónsson function for λ is a function $f\colon [\lambda]^\omega \to \lambda$ [2]with the property for each $x \subseteq \lambda$ such that $\overline{\overline{x}} = \lambda, f''[x]^\omega = \lambda$.

Lemma 1.2 (Erdős–Hajnal, 1966)**.** *For every infinite ordinal λ, there is an ω-Jónsson function for λ.*

Proof. [3] For $x, y \in [\lambda]^\omega$, define an equivalence relation $\sim$, we say $x \sim y$ if there exists an ordinal $\alpha < \bigcup x$ such that $x - \alpha = y - \alpha$. Let E be the equivalence class of x and choose a representative

[1] The implication $\mathrm{Con}(\mathsf{NBG}) \Rightarrow \mathrm{Con}(\mathsf{ZFC})$ is immediate, whereas the converse can be established using class forcing.

[2] Here $[\lambda]^\omega = \{A \subseteq \lambda\colon \overline{\overline{A}} = \aleph_0\}$.

[3] This proof is adapted from Andrés Caicedo, *580 – Partition Calculus (3)*. https://caicedoteaching.wordpress.com/2009/04/06/580-partition-calculus-3/

x_E from each $\sim$-equivalence class E(here we need AC!), then let $\alpha \in x_E$ be the least ordinal such that $x_E - (\alpha + 1) = x - (\alpha + 1)$ and consider the function $f(x) = \alpha$.

Claim: there exists $A \subseteq \lambda$ of size λ such that $\forall B \subseteq A$ of size $\lambda, f''[B]^\omega \supseteq A$.

Now we prove this claim: assume to the contract, such an A does not exist. By induction, we can construct a descending chain $(A_n \colon n < \omega)$ together with a sequence $(a_n \colon n < \omega)$ such that for each $n < \omega, A_{n+1} \in [A_n]^\lambda$ and $a_{n+1} \in A_n \backslash (\alpha_n + 1)$ such that $a_{n+1} \notin f''[A_{n+1}]^\omega$.

Setting $\lambda = A_0$. Suppose $A_n \in [\lambda]^\lambda$ and $a_n < \lambda$ are already constructed. Now consider $C := A_n \backslash (a_n + 1)$, and $\overline{\overline{C}} = \lambda$ as λ is an infinite ordinal. Apple the negation of the claim, we find $A_{n+1} \in [C]^\lambda$ and $a_{n+1} \in C$ such that $a_{n+1} \notin f''[A_{n+1}]^\omega$.

Set $y := \{a_1, a_2, \dots\} \in [\lambda]^\omega$, let F be the equivalence class of y and say $x_F \sim y$. Then there exists an $m < \omega$ and an ordinal α such that $\{a_n \colon m \leq n < \omega\} = x_F - \alpha$. Let $y' := \{a_n \colon m < n < \omega\} \subseteq y$, but in fact, $y' = x_F - (a_m + 1)$. Thus $f(y') = a_m, y' \subseteq A_m$, so $y' \in [A_m]^\omega \implies a_m = f(y') \in f''[A_m]^\omega$, contradiction! Hence, we have proved the claim, which yields the lemma by transporting $f \restriction [A]^\omega$ along a bijection between λ and A. ■

Now we go back to the proof of Kunen's Inconsistency.

Theorem 1.3 (Kunen)**.** *If there exists a non-trivial elementary embedding $j \colon V \to M$ then $V \neq M$.*

Proof. Let λ be the critical supremum of j. Assume $j''\lambda \in M$. Let $f \in V$ be an ω-Jónsson function for λ, then $j(f) \in M$ and by elementarity, $M \vDash$ "$j(f)$ is an ω-Jónsson function for $j(\lambda)$". Since $j''\lambda \subseteq \lambda$ and $\overline{\overline{j''\lambda}} = \lambda$ so $j(f)''[j''\lambda]^\omega = \lambda$ in M. Consider $j(f)(s)$ where $s \in [j''\lambda]^\omega$: there eixsts $A = \{\alpha_0, \alpha_1, \dots\} \in [\lambda]^\omega$ such that $s = j''A = \{j(\alpha_0), j(\alpha_1), \dots\}$. Look at $j(A)$: for each $n \leq \omega, j(n) = n$, thus $j(A)(n) = j(A(n)) = j(\alpha_n)$, in fact $j(A) = j''A$. Therefore we have $j(f)(s) = j(f)(j(A)) = j(f(A))$, the first equality is given by $j''A = j(A)$ and the second is given by elementarity. Since $f(A) \in \lambda$ thus $j(f(A)) \in j''\lambda$. But $j(f)''[j''\lambda]^\omega \subseteq j''\lambda \subsetneq \lambda$ which contradicts $j(f)''[j''\lambda]^\omega = \lambda$. ■

1.2 Woodin's proof

This proof only depends on partitioning stationary sets into disjoint stationary subsets. Let us introduce the main theorem that will be used in the proof.[4]

Recall:

Definition 1.4 (Diagonal Intersection)**.** Let α be regular, and let $X_\xi \subseteq \alpha$ for all $\xi < \alpha$. The diagonal intersection of $\{X_\xi, \xi < \alpha\}$, abbreviated by $\Delta_{\xi<\alpha} X_\xi = \{\eta < \alpha \colon \eta \in \bigcap_{\xi<\eta} X_\xi\}$.

Definition 1.5. Let α be regular, and let F be a filter on α, F is called **normal** if and only if for all $\{X_\xi \colon \xi < \alpha\} \subseteq F, \Delta_{\xi<\alpha} X_\xi \in F$.

Lemma 1.6. *Let α be regular, and let $F_\alpha = \{A \subseteq \alpha \colon \exists B \supseteq A, B$ is a club$\}$ be the club filter on α, then F_α is normal.*

[4]This subsection is largely based on Chapter 4.2 of Prof. Schindler's Set Theory book.

Theorem 1.7 (Fodor)**.** *Let α be uncountable and regular, and let $S \subseteq \alpha$ be stationary. Let $f\colon S \to \alpha$ be regressive in the sense that $f(\xi) < \xi$ for all $\xi \in S$. Then there is a stationary $T \subseteq S$ such that $f \restriction T$ is constant.*

The following theorem is the key tool used in Woodin's proof:

Theorem 1.8 (Solovay)**.** *Let κ be a regular uncountable cardinal, and let $S \subseteq \kappa$ be stationary. Then S may written as a disjoint union of κ stationary sets, i.e., there is $(S_\xi\colon \xi < \kappa)$ such that $S_\xi \subseteq S$ is stationary in κ for every $\xi < \kappa, S_\xi \cap S_{\xi'} = \emptyset$ for all $\xi, \xi' < \kappa$ with $\xi \neq \xi'$, and $S = \bigcup_{\xi<\kappa} S_\xi$.*

Proof. Let us write $S = S_0 \biguplus S_1$ where $S_0 = \{\alpha \in S\colon \operatorname{cf}(\alpha) < \alpha\}, S_1 = \{\alpha \in S\colon \operatorname{cf}(\alpha) = \alpha\}$, at least one of S_0, S_1 must be stationary.

Claim: There is some stationary $\bar{S} \subseteq S$ and some sequence $(A_\alpha\colon \alpha \in \bar{S})$ such that $A_\alpha \subseteq \alpha$ is a club in α and $A_\alpha \cap \bar{S} = \emptyset$ for every $\alpha \in \bar{S}$.

Now we prove this claim: suppose S_0 is stationary, apply Fodor's theorem, there is some stationary $\bar{S} \subseteq S_0$ and some regular $\lambda < \kappa$ such that $\operatorname{cf}(\alpha) = \lambda$ for all $\alpha \in \bar{S}$. Pick $(A_\alpha\colon \alpha \in \bar{S})$ where A_α is club in $\alpha, \operatorname{otp}(A_\alpha) = \lambda$ and $\operatorname{cf}(\delta) < \lambda$ for every $\delta \in A_\alpha$, thus $A_\alpha \cap \bar{S} = \emptyset$ for all $\alpha \in \bar{S}$. Now suppose S_1 is stationary. Let $\bar{S} = \{\alpha \in S_1\colon S_1 \cap \alpha \text{ is non-stationary}\}$. We must have that $\bar{S}$ is stationary. To see this, let $C \subseteq (\kappa \backslash (\omega+1))$ be club, and let $C' \subseteq \kappa$ be the club of all limit points of C. If $\alpha = \min(S_1 \cap C')$, then $C \cap \alpha$ is club in α so $C' \cap \alpha$ is still club in α, as $\alpha > \omega$ is regular; but $(C' \cap \alpha) \cap S_1 = \emptyset$, so $\alpha \in C \cap \bar{S}$. Thus we may pick $(A_\alpha\colon \alpha \in S)$ where A_α is a club in α and $A_\alpha \cap \bar{S} \subseteq A_\alpha \cap S_1 = \emptyset$ for each $\alpha \in \bar{S}$. Hence we proved the claim.

For $\alpha \in \bar{S}$, let $(\gamma_i^\alpha\colon i < \operatorname{otp}(A_\alpha))$ be the monotone enumeration of A_α.

Claim: There exists $i < \kappa$ such that for all $\beta < \kappa, \{\alpha \in \bar{S}\colon i < \operatorname{cf}(\alpha) \wedge \gamma_i^\alpha > \beta\}$ is stationary in κ.

Now we prove this claim: suppose $\bar{S} \subseteq S_0$ and let $\lambda < \kappa$ be such that $\operatorname{cf}(\alpha) = \lambda$ for every $\alpha \in \bar{S}$. Assume to the contract, for every $i < \lambda$, there is some $\beta_i < \kappa$ and some club $C_i \subseteq \kappa$ such that $\forall \alpha \in \bar{S} \cap C_i, \gamma_i^\alpha \leq \beta_i$. But then if $\tilde{\beta} = \sup(\{\beta_i\colon i < \lambda\}) < \kappa$ and $\alpha \in (\bar{S} \cap \bigcap_{i<\lambda} C_i) \backslash (\tilde{\beta}+1)$, then $\gamma_i^\alpha \leq \beta_i$ for all $i < \lambda$, since $\operatorname{otp}(A_\alpha) = \lambda$ for all $\alpha \in \bar{S}$, so A_α would be bounded in α. Contradiction! Now suppose $\bar{S} \subseteq S_1$, so $\operatorname{cf}(\alpha) = \alpha$ for all $\alpha \in \bar{S}$. Again, assume to the contract, then for every $i < \kappa$ there is some $\beta_i < \kappa$ and some club $C_i \subseteq \kappa$ such that for all $\alpha \in C_i \cap \bar{S}$, either $i \geq \operatorname{cf}(\alpha)$ or $\gamma_i^\alpha \leq \beta_i$. Let $D \subseteq \kappa$ be the club of all $\beta < \kappa$ such that $i < \beta$ implies $\beta_i < \beta$. By 1.6 we may pick $\alpha, \alpha' \in \bar{S} \cap \Delta_{i<\kappa} C_i \cap D$, with $\alpha < \alpha'$. If $i < \alpha$, then $\alpha' \in \bar{S} \cap C_i$, so that $\gamma_i^{\alpha'} \leq \beta_i$, but $\alpha \in D$, so $\beta_i < \alpha$. Hence $\gamma_i^{\alpha'} < \alpha$ for all $i < \alpha$, which implies $\gamma_\alpha^{\alpha'} \leq \alpha$. by closedness of $A_{\alpha'}$. On the other hand, clearly $\gamma_i^{\alpha'} \geq i$ for all $i \leq \alpha'$, so in fact, $\gamma_\alpha^{\alpha'} = \alpha$, then $\gamma_\alpha^{\alpha'} \in \bar{S}$, contradiction!

Now we proved the claim, fix $i_0 < \kappa$ such that $\forall \beta < \kappa, \{\alpha \in \bar{S}\colon i_0 < \operatorname{cf}(\alpha) \wedge \gamma_{i_0}^\alpha > \beta\}$ is stationary in κ. Let us recursively define stataionary sets S_ξ and ordinals β_ξ for $\xi < \kappa$ as follows: Fix $\xi < \kappa$, and suppose that $S_{\bar{\xi}}$ and $\beta_{\bar{\xi}}$ have already been chosen for all $\bar{\xi} < \xi$. The set $\{\alpha \in \bar{S}\colon \gamma_{i_0}^\alpha > \sup_{\bar{\xi}<\xi} \beta_{\bar{\xi}}\}$ is stationary in κ. Let we use Fodor's theorem again (as the function $\alpha \to \gamma_{i_0}^\alpha$ is regressive) to pick some $\beta > \sup_{\bar{\xi}<\xi} \beta_{\bar{\xi}}$ and some stationary $S^* \subseteq \bar{S}$ such that for all $\alpha \in S^*, \gamma_{i_0}^\alpha = \beta$. Let us set $S_\xi = S^*$ and $\beta_\xi = \beta$.

It is clear that $S_\xi \cap S_{\xi'} = \emptyset$ whenever $\xi \neq \xi'$. Set $T_0 = S_0 \cup \left(S \setminus \bigcup_{\xi<\kappa} S_\xi\right)$[5] and $T_\xi = S_\xi$ for all $0 < \xi < \kappa$. Then $S = \biguplus_{\xi<\kappa} T_\xi$, where the T_ξ's are pairwise disjoint stationary subsets of κ.

■

Finally, we can return to Woodin's proof:

Theorem 1.9 (Kunen)**.** *There is no non-trivial elementary embedding $j\colon V \to V$.*

Proof. Let $\kappa = \kappa_0$ be the critical point of j and let λ be the critical supremum of j, we know $j(\lambda) = \lambda$ thus $\lambda^+ = j(\lambda)^+ = j(\lambda^+)$. ■

[5]Here S_0 denotes the first stationary set we inductively constructed rather than the set $\{\alpha \in S\colon \operatorname{cf}(\alpha) < \alpha\}$ we defined in the beginning, I know it is abuse of notation, but it is what it is.